

REVIEW

Discordant non-invasive prenatal testing (NIPT) – a systematic review

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ABSTRACT

With a high sensitivity and specificity, non-invasive prenatal testing (NIPT) is an incomparable screening test for fetal aneuploidy. However, the method is rather newly introduced, and experiences with discordant results are few. We did a systematic review of literature reporting details of false positive and false negative NIPT results. Discordant sex chromosome results were not included. We identified 22 studies reporting case details. In total, 206 discordant cases were included, of which 88% were false positive and 12% false negative. Details on maternal age, gestational age, platform/company, Z-score, fetal fraction, results and explanation were specified. The main reasons for discordant results were confined placental mosaicism, maternal copy number variation, vanished twin, maternal cancer and true fetal mosaicism. A very high percentage of cases (67%) were reported with no obvious biological or technical explanation for the discordant result. The included cases represent only a minor part of the true number of false positive or false negative NIPT cases identified in fetal medicine clinics around the world. To ensure knowledge exchange and transparency of NIPT between laboratories, we suggest a systematic recording of discordant NIPT results, as well as a quality assurance by external quality control and accreditation. © 2017 John Wiley & Sons, Ltd.

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INTRODUCTION

Non-invasive prenatal testing (NIPT) represents a paradigm shift in fetal medicine. In 1997, cell-free fetal DNA fragments were discovered in maternal blood,¹ and in 2011, NIPT was introduced into clinical practice. NIPT by massively parallel sequencing of cell-free fetal DNA (cffDNA) fragments has rapidly become common in prenatal screening. Many thousands of pregnant women have now undergone NIPT to identify the potential presence of trisomy 21, 18 or 13 in their fetus.

Several studies validating NIPT have been published, and the method is well evidenced for detecting trisomies 21, 18 and 13. In a meta-analysis, the detection rate of T21 was found to be 99.2% with a 0.09% false positive rate.² Nevertheless, NIPT is not considered as a diagnostic test, as there is a small chance of a false positive or a false negative result. The main reason that discordant results occur is because the circulating cffDNA is mainly derived from the cytotrophoblasts of chorionic villi in placenta,^{3,4} which is not always representative for the fetus. Discrepancy between placental and fetal karyotype and the extent of mosaicism can compromise the NIPT result. Accordingly, a positive NIPT result should always be confirmed by an invasive test (chorion villus sampling or amniocentesis).

With a high sensitivity and specificity, NIPT is incomparable as a screening test for trisomies 21, 18 and 13. However, the

method is rather newly introduced, and experiences with discordant results are few. We wanted to review published studies involving discordant NIPT results with the aim to investigate characteristics of false positive and false negative NIPT cases. Knowledge about the limitations of NIPT and reasons for discordant results are fundamental for professionals responsible for analysis, as well as pre-test and post-test counselling.

MATERIAL AND METHODS

We performed a systematic review of literature published between 1 January 1997 and 1 March 2016 in the online database PubMed. The search combination used was (NIPT [all fields] OR (Non-Invasive Prenatal Testing) OR (cell free fetal DNA)) AND (False positive[all fields] OR False negative[all fields] OR discordant[all fields]). In addition, reference lists and suggestions from colleagues were checked for further discordant NIPT cases. Only manuscripts written in English were included.

Studies were included if they reported details for at least one case of a false positive or false negative NIPT result of any autosomal chromosome. Discordant sex chromosome cases were not included.

The search on PubMed identified 229 papers. Screening of headlines and abstracts excluded 176 papers as they obviously did not fit the inclusion criteria. A total of 53 manuscripts were read, and 16 were found to include characteristics of at least one false positive or false negative case. Additionally, six papers were identified through reference review and recommendations. The 22 studies included, dating from 2013 to 2016, describe in total 206 discordant cases.

We compared available biological and technical characteristics of false positive and false negative cases. The following parameters are reported in Tables 1 and 2: maternal age, gestational age when NIPT was performed, platform/company, Z-score, fetal fraction (FF) (and effective FF), results and explanation. We wanted to include sequencing depth, but none of the included papers did report on this.

The NIPT findings indicating a high-risk result in presence of an euploid fetus were categorized as false positive. NIPT findings indicating a low-risk result in presence of an abnormal fetal karyotype or a high-risk result for a chromosome abnormality different from the one present in the fetus were categorized as false negatives, even though there might have been a biological explanation in the placenta. The nomenclature 'false positive' and 'false negative' refers to discordance between the NIPT result and fetal karyotype, thus as from a pregnant woman's point of view.

RESULTS

Characteristics of all identified cases are listed in Tables 1 and 2. We identified characteristics of 182 false positive cases (88%) and 24 false negative cases (12%). Totally, 133 cases (65%) were reported with no obvious biological or technical explanation for the discordant result. In 28 cases (14%), the FF was not reported. All the reported FFs were above 4%.

False positive cases

Of all reported false positive cases, 65 euploid cases (36%) had an NIPT positive for T21,^{10,12,16} 59 euploid cases (32%) were predicted to have T18,^{7,8,10,12,14} 47 cases (26%) were predicted to have T13,^{7,8,11,12} one case (0.5%) was predicted to have T22⁶ and one (0.5%) T9.¹⁵ One case (0.5%) had an NIPT result indicating monosomy 18.¹¹ Eight cases (4%) had an NIPT result showing multiple aneuploidies, whereof one trisomy 21 case had an NIPT result predicting multiple aneuploidies explained by placental mosaicism with two different trisomy cell lines.¹³ In 122 false positive cases (67%), there was no explanation for the discordant result.^{7,8,10,12,14} Among the 60 false positive cases with a biological or technical explanation (33%), 29 (48%) was found to have maternal copy number variation (CNV),^{12,14} and 19 (32%) confined placental mosaicism (CPM).^{6,7,10,12,13,15} In one case (2%), the explanation was found in a T21 positive vanished twin,⁵ and in another case (2%), the mother turned out to have a supernumerary ring derived from chromosome 18 in 35% of her cells (maternal mosaicism).⁸ One case (2%) was described with a maternal and fetal chromosome 21 duplication causing the false positive result.¹⁶ Nine cases (15%) were reported as false positive cases because of maternal cancer.^{11,17} Six of these cases had an NIPT result

indicating multiple aneuploidies.^{11,17} The results of false positive cases are summarized in Figure 1.

False negative cases

Among the false negative cases ($n = 24$), 10 cases (42%) turned out to have T21 or T21 mosaicism,^{12,18,22,24} nine cases (38%) had T18 or T18 mosaicism^{10,12,26} and three cases (13%) had T13 or T13 mosaicism.^{19,22,26} One false negative case (4%) had an NIPT result indicating 47, XXX, but the fetal karyotype was found to be 48, XXX, +18.²⁰ Placenta biopsies from this case had 20% to 30% of the cells with karyotype 48, XXX, +18, while the remaining 70% to 80% had karyotype 47, XXX. Another case had a T21 positive NIPT result with a chorionic villus sampling (CVS) showing mosaicism of T21 and 48, XY, +18, +21 in 45% of examined cells.²² In one of the false negative cases with an NIPT result indicating T21, the fetal karyotype was found to be T18, but with placental mosaicism of both T18, T21 and euploid cell lines.²¹ True fetal mosaicism (TFM) was the explanation for another two false negative cases reported with a 45, X positive NIPT result but T18 positive fetuses.^{12,23} Additionally, one false negative case had an NIPT result indicating T13, with a T18 positive CVS.²⁵ In a case report of a false negative T18 case with a normal NIPT result, the explanation of the discordant result was probably explained by incorrect clinical annotation.²⁷ In three of the unexplained false negative cases, the FF was not reported and might have been lower than 4%. Two of these cases did neither report the gestational age. The results of false negative cases are summarized in Figure 1.

DISCUSSION

A large number of validation studies reporting sensitivity and specificity of NIPT have been published, but studies reporting discordant NIPT results are few compared with the large number of performed NIPT analyses. This review does not represent all discordant cases because of publication bias and an incomplete reporting of cases. Professionals working with NIPT should be aware of reasons for false positive and false negative test results, both to ensure comprehensive patient information and to illuminate future points of improvement. Numerous studies have shown sensitivity rates of approximately 99% with false positive rates below 1% for NIPT.² However, the positive predictive value is limited to 40% to 90%.²⁸ This emphasizes the need for knowledge about the mechanisms underlying false positive and false negative results and transparency when it comes to discordant cases.

Cases of pregnant women terminating their pregnancy after a positive NIPT result but with no prenatal karyotype confirmation have been published. Unconfirmed interruption of pregnancy has been reported in 6.2% to 19.6% of cases with a positive NIPT result.^{29,30} Dobson *et al.*²⁹ found that 4% of 114 women in United States who received a positive NIPT result but did not have sonographic findings chose termination without confirmed prenatal karyotype. Based on these findings pre-test and post-test counselling, including information about the risk of a false positive result, should be mandatory for all women choosing NIPT.

Table 1 Biological and technical characteristics of false positive cases

Reference	Maternal age	Gestational age	Platform/company	Z-score	Fetal fraction	Results	Explanation
Osborne <i>et al.</i> ¹⁷	37	13	Verinata	NCV Chr 13 = 10.9 NCV Chr 18 = -10.7	n.r.	NIPT[GA 13 + 17]: T13, monosomy 18 NIPT [GA36]: 'nonreportable' (indicates monosomy 13 and 18) Fetus: 46, XY	Maternal malignancy
				NCV Chr 13 = 17.6 NCV Chr 18 = -20.8		Placenta: normal Maternal neuroendocrine carcinoma (vaginal origin)	
				'Nonreportable' Significant reduced for 13 and 18			
Choi <i>et al.</i> ⁶	39	20 + 3		Chr 22 = 6.92	n.r.	NIPT: T22 Placenta: T22 Umb. cord: normal Fetus: normal	CPM
Dugo <i>et al.</i> ⁷							
Case 1	40	<13	n.r.	n.r.	n.r.	NIPT: T13 CVS: 46, XX	Unknown
Case 2	44	<12	n.r.	n.r.	n.r.	NIPT: T18 CVS: 46, XX/47, XX, +18 AC: 46, XX	CPM
Clark-Ganheart ⁸							
Case 1	23	19+	n.r.	Chr 18 = high risk	n.r.	NIPT: trisomy 18 AC: normal Mother: 35% mosaicism of a supernumerary ring derived from chr 18	Maternal mosaicism
Case 2	26	20+	n.r.	Chr 13 = high risk	n.r.	NIPT: trisomy 13 AC: chr 8p duplication/deletion (8.1 Mb)	Unknown
Grömminger <i>et al.</i> ⁵	39	17 + 2 [two samples]	Illumina HiSeq 2000	Chr 13 = 1.3/0.4, Chr 18 = -1.5/-1.7, Chr 21 = 13.5/11.1	20.7%/24.8% Y-chrom rep = 9.2%/9.3%	NIPT[GA 17]: trisomy 21 NIPT [GA 38]: normal AC: 46, XY Vanished twin: 47, XX, +21	Vanished twin
		38 + 2		Chr 13 = 0.8, Chr 18 = 1.9, Chr 21 = -0.3	21.4% Y-chrom rep = 21.7%	Newborn: 46, XY Placenta [samples]: 46, XY Mother: 46, XX	

(Continues)

Table 1 (Continued)

Reference	Maternal age	Gestational age	Platform/company	Z-score	Fetal fraction	Results	Explanation
Lebo <i>et al.</i> ¹⁰ Case 1	9 + 3 23	n.r.	MaterniT21 (Sequenom)	n.r.	n.r.	NIPT (GA 9): T18 NIPT (GA 23): normal Fetus: 46, XY Placenta biopsies: 46, XY	Unknown
						NIPT: T21 AC: 46, XY Newborn: 46, XY	Unknown
						NIPT: T21 AC: 46, XY Placenta: 47, XY,+21(18)/46, XY ² Newborn: 46, XY	CPM
Case 2	16 + 5	n.r.	Harmony (Ariosa Diagnostics)	n.r.	n.r.		
Case 3	n.r.	n.r.	Harmony (Ariosa Diagnostics)	n.r.	n.r.		
Bianchi <i>et al.</i> ¹¹ Case 1	37	13	Verifi Prenatal Test (Illumina)	n.r.	n.r.	NIPT: monosomy 18, T13 Fetus: 46, XY Maternal cancer (neuroendocrine)	Maternal malignancy
						NIPT: monosomy 18 Fetus: not performed Newborn: female Maternal non-Hodgkin (B-cell lymphoma)	Maternal malignancy
						NIPT: trisomy 13 Fetus: 46, XY Maternal colorectal cancer Postnatal maternal DNA sequencing: T13, monosomy X	Maternal malignancy
						NIPT: monosomies 13, 18 and 21 Fetus: 46, XY Maternal Hodgkin lymphoma Postnatal maternal DNA sequencing: Monosomies 13, 18, 21 and X	Maternal malignancy
						NIPT: trisomies 21 and 13, monosomy 18 Fetus: 46, XY Maternal Acute T-cell lymphoblastic leukaemia Maternal preeclampsia	Maternal malignancy

(Continues)

Table 1 (Continued)

Reference	Maternal age	Gestational age	Platform/company	Z-score	Fetal fraction	Results	Explanation
Case 6	37	12				NIPT: trisomy 18 Fetus: 46, XX Maternal non-Hodgkin (B-cell) lymphoma	Maternal malignancy
Case 7	39	11				NIPT: monosomy 18, XXY Fetus: 46, XY Maternal Non-Hodgkin (B-cell) lymphoma	Maternal malignancy
Case 8	39	10				NIPT: trisomies 13, 18 and 21, monosomy X Fetus: 46, XX Maternal anal cancer	Maternal malignancy
Zhang <i>et al.</i>¹² 157 false positive T13, T18 or T21 cases	Mean = 31.65 (range 21–44)	Mean = 18.5 (range 13–27)	Illumina HiSeq 2000	n.r.	Mean = 9.74% (range 3.54–21.94)	61 cases: NIPT: T21 15 cases with suspected maternal CNV (0.59–6.72 Mb) Three cases with suspected mosaicism/CPM Two twin pregnancies with normal karyotype of both fetuses 51 cases: NIPT: T18 Nine cases with suspected maternal CNV (0.5–3.35 Mb) Two cases with suspected mosaicism/CPM 44 cases: NIPT: T13 Three cases with suspected maternal CNV (4.89–14 Mb) Nine cases with suspected mosaicism/CPM One case with multiple chromosomal aneuploidy (chr 13, chr 14, chr 20) In total 116 cases with no obvious biological or technical explanation.	Maternal CNV Mosaicism/CPM Unknown Maternal CNV Mosaicism/CPM Maternal CNV Mosaicism/CPM Unknown

(Continues)

Table 1 (Continued)

Reference	Maternal age	Gestational age	Platform/company	Z-score	Fetal fraction	Results	Explanation
Crooks <i>et al.</i>¹³	32	34	Verifi by progenity	Chr 18 and 21 = Aneuploidy detected Chr 13 = Aneuploidy not detected	n.r.	NIPT: T18 and T21 Newborn: 47, XY, +21 Placenta: 48, XY, +18, +21 (49/800 cells), 47, XY, +21 (751/800 cells)	CPM
Snyder <i>et al.</i>¹⁴ Case 1	36	18	Illumina Verifi	Chr 18 = high risk	n.r.	NIPT: T18 AC: 46, XY Newborn: healthy male Mother: CNV, 1.15 Mb duplication on chromosome 18	Maternal CNV
Case 2	25	20		Chr 18 = high risk	n.r. Maternal BMI = 65	NIPT: T18 Newborn: normal exam	Unknown
Case 3	34	12		Chr 18 = high risk	n.r.	NIPT: T18 Newborn: normal exam Mother: CNV, 487 kb	Maternal CNV
Case 4	38	12		Chr 13 = high risk	n.r.	NIPT: T18 AC: 46, XX Newborn: normal exam	Unknown
Cheng <i>et al.</i>¹⁵	40	32	NIFTY-Extended test, BGI	Chr 13, 18 and 21 = low risk Chr 9 = high risk		NIPT: T9 CVS: 47, XX, +6 [20/20], 47, XX, +21 [7/20] Cord blood: 46, XX Placenta: 47, XX, +6	CPM
		32+	GWNS-test (private lab)	Chr 9 = high risk	13% (4.6% EF for chromosome 21)		
Meschino <i>et al.</i>¹⁶	37	21	MaterniT21 (Squenom)	Chr 21 = high risk	n.r.	NIPT: T21 Mother: 46, XX, partial chromosome 21 duplication [21q21.1] – including gene for early onset Alzheimer Fetus: partial chromosome 21 duplication [21q21.1] – including gene for early onset Alzheimer	Maternal and fetal chromosome 21 duplication

n.r., not reported; EF, effective fraction; CPM, confined placental mosaicism; CNV, copy number variation; NCV, normalized chromosome value (>4, affected chromosome; <2.5, unaffected; 2.5–4.0, unclassified); GWNS, genome-wide normalized score; CVS, chorionic villus sampling; TFM, true fetal mosaicism.

Table 2 Biological and technical characteristics of false negative cases

Reference	Maternal age	Gestational age	Platform/company	Z-score	Fetal fraction	Result	Explanation
Wang <i>et al.</i>¹⁸ Case 1 IVF	32	18 (two samples)	n.r.	Chr 21 = 2.04/1.77	15.55% (EF = 3%)	NIPT [GA 18]: normal NIPT [GA 24]: T21 mosaicism Cordocentesis: 46, XX, der(21;21)(Q10:Q10), +21 Placenta: 17–53% T21 mosaicism Umb. cord: 90% T21 mosaicism	TFM
		24 (two samples)		Chr 21 = 4.0/3.49			
Case 2	35	18	n.r.	Chr 21 = 1.33	19.72% (EF < 4.0%)	NIPT: normal AC: 47, XY, +21 Placenta: 2.3–51% mosaicism Fetus: >92% T21	TFM
Canick <i>et al.</i>²² Case 1	34	14	Sequenom	Chr 13 = 0.08	6% [0.6%EF]	NIPT: normal AC: 46, XX/47, XX, +13 (10%)	TFM
				Chr 21 = 2.03 Chr 18 = 0.21 Chr 13 = -0.96	17% [1.7%EF]	NIPT: normal CVS: 46, XY/47, XY, +21 (10%)	TFM
Case 3	39	12		Chr 21 = 12.71 Chr 18 = -0.22 Chr 13 = -0.40	23% [10%EF]	NIPT: T21 CVS: 47, XY, +21/48, XY, +18 + 21[45%]	TFM
Gao <i>et al.</i>²⁰	43	13 + 2	Illumina HiSeq 2000	Chr 13, 18 and 21 = low risk XXX = high risk	7.4% [EF < 2.2%]	NIPT: XXX AC: 48, XXX, +18 Placenta: 48, XXX, +18 (20–30%)/47, XXX Fetus: 48, XXX, +18	TFM
Mao <i>et al.</i>²¹	22	17	Illumina HiSeq 2000	Chr 21 = 4.38 Chr 18 = 0.35	n.r.	NIPT [GA 17]: T21 NIPT [GA 16]: T21 AC: 47, XX, +18	TFM
		19		Chr 21 = 8.05 Chr 18 = -0.7	GA 19 = 11.55%	Fetus: T18 Placenta: 49, 8% T21, 34, 8% T18, 15, 4% euploid cells	
Pan <i>et al.</i>²³	24	17 + 4	Illumina HiSeq 2000	Chr 18 = -0.52 X = -4.05	n.r.	NIPT: 45, X AC: 47, XX, +18 Fetus: 47, XX, +18 Placenta: 45, X (67%)/47, XX, +18 (30%)	TFM
Smith <i>et al.</i>²⁴	33	n.r.	Verif by Progenity	n.r.	n.r.	NIPT: normal Newborn: 47, XX, +21	Unknown

(Continues)

Table 2 (Continued)

Reference	Maternal age	Gestational age	Platform/company	Z-score	Fetal fraction	Result	Explanation
Sepulveda²⁵	36	12 + 2	Harmony (Ariosa diagnostics)	Chr 13 = high risk (>99%) Chr 18 and 21 = low risk (<0.01%)	9.1%	NIPT: T13 CVS: 47, XY, +18	Unknown
Hochstenbach <i>et al.</i>²⁶							
Case 1	35	13 + 5	n.r.	Chr 13, 18 and 21 = low risk	GA13 = 8.8%	NIPT (GA 13): normal NIPT (GA 20): T13 AC: 47, XY, +13 Fetus: 47, XY, +13 Placenta + umb. Cord: 47, XY, +13 (96–97%)	Unknown
Case 2	40	11	n.r.	Chr 13, 18 and 21 = low risk	GA11 = 10.7%	NIPT (GA 11): normal NIPT (GA 19): T18 AC: 47, XX, +18 Placenta: 47, XX, +18 (70–80%)	TFM
Stokowski <i>et al.</i>²⁷	n.r.	n.r.	Ariosa Diagnostics	n.r.	n.r.	NIPT: normal Fetal karyotype: 47, XY, +18 Fetal tissue: 46, XY	Probably incorrect clinical annotation
Lebo <i>et al.</i>¹⁰							
		n.r.	MaterniT21 (Sequenom)	n.r.	n.r.	NIPT: normal AC: 47, XX, +18	Unknown
Zhang <i>et al.</i>¹²							
Case 1	26	19	Illumina HiSeq 2000	n.r.	9.9%	NIPT: normal New born: 47, XY, +21	Unknown
Case 2	28	22			12.9%	NIPT: normal New born: 47, XY, +21	Unknown
Case 3	26	25			13.4%	NIPT: normal New born: 46, XY delet(21:21)(q10;q10)+21	Unknown
Case 4	22	14 + 4			n.r.	NIPT: normal New born = 47, XX, +21	Unknown
Case 5	28	25 + 4			10.9%	NIPT = normal New born: 47, XX, +21	Unknown
Case 6	37	13			13.1%	NIPT: normal Newborn: 47, XX, +21	Unknown
Case 7	27	17 + 3			5.2%	NIPT: normal Fetus: 47, XY, +18	Unknown

(Continues)

Table 2 (Continued)

Reference	Maternal age	Gestational age	Platform/company	Z-score	Fetal fraction	Result	Explanation
Case 8	29	19 + 4			5.3%	NIPT: normal Fetus: 47, XY, +18 Placenta: 20–40% mosaicism T18 and tetraploidy in 6 places	TFM
Case 9	24	20 + 4			9.5%	NIPT: 45, X AC: 47, XX, +18 Fetus: 47, XX, +18 Placenta: T18 (30%), 45, X (60%)	TFM
Cao <i>et al.</i> ¹⁹	19	20	n.r.	Chr 13, 18 and 21 = low risk	8.5% (EF < 4%)	NIPT: normal Newborn: 47, XY, +13 Placenta: 32–64% T13 mosaicism	TFM

n.r., not reported; EF, effective fetal fraction; CPM, confined placental mosaicism; CNV, copy number variation; CVS, chorionic villus sampling; TFM, true fetal mosaicism.

MOSAICISM

As found in this review, CPM is one of the major factors causing false positive cases, here seen in 32% of all reported false positive cases. Accordingly, TFM type 5 is the main reason for false negative cases, here described in 92% of all reported false negative cases. CffDNA originate from chorionic villi in the placenta, where the constant turnover of cytotrophoblasts induces apoptosis and release of cffDNA to maternal blood. From the introduction of CVS, we know that the cytotrophoblast layer of chorionic villi is not always representative for the fetus, because it is derived from the trophoblast of the blastocyst, whereas the fetus is derived from the inner cell mass (epiblast).³¹ Postzygotic mitotic division errors in chromosomally normal and abnormal conceptuses will lead to chromosomal mosaicism, which is found in about 1% to 2% of chorionic villi samples.^{32–34} The karyotypes of the cytotrophoblast, the mesenchymal core and the fetus may be different. Most often, mosaicism is characterized as CPM where the abnormal cell line is confined to the cytotrophoblast and/or the mesenchymal core of the chorion villi, whereas the fetus has a normal karyotype. CPM accounts for 87% of chorionic villi mosaicism cases.³⁴ If the abnormal cells are only restricted to the cytotrophoblast layer, and not present in the mesenchymal core or fetus, it is categorized CPM type 1 and will possibly cause a false positive NIPT result dependent on the extent of mosaicism. In CPM type 3, the abnormal cells are located in both layers of the chorion villi and absent in the fetus and might also cause a false positive result. In CPM type 2, the abnormal cells are confined to the mesenchyme only, and the NIPT result will correspond to the fetal karyotype. Cases of TFM type 5 (TFM 5), which is associated with a normal karyotype in the cytotrophoblast layer while the fetus itself has an abnormal karyotype, will cause a false negative NIPT result.⁹ As placental mosaicism is a relatively common biological phenomenon (1–2% of pregnancies), NIPT is, and will continue to be, a screening test where discordant cases will occur.

VANISHING TWIN

Another phenomenon contributing to discordant cases is vanishing twins, as described by Grömminger *et al.*⁵ They report a false positive case, where NIPT indicated trisomy 21, but amniocentesis, placenta, mother and the new born baby were found to be euploid. The explanation for the discordant NIPT result was found in a vanished twin identified by ultrasound in the first trimester. At birth of the surviving twin, a fetus papyraceus was found in the placenta, which was confirmed to be a trisomy 21 positive female (47, XX, +21). An additional NIPT analysis in gestational week 38 + 2 had no indications of trisomy, as the deceased twin seemed to no longer release cffDNA to maternal bloodstream. For how long a placenta of a vanished twin can release cffDNA to maternal blood is unknown, but Curnow *et al.* have reported on five studied pregnancies where cffDNA from a vanished twin was detectable for up to 8 weeks following co-twin demise.³⁵ Studies regarding the frequency of vanishing twin syndrome are few, but Thurik *et al.* estimated a frequency of vanishing twins of at least around 0.23% of pregnancies.³⁶ Cases with

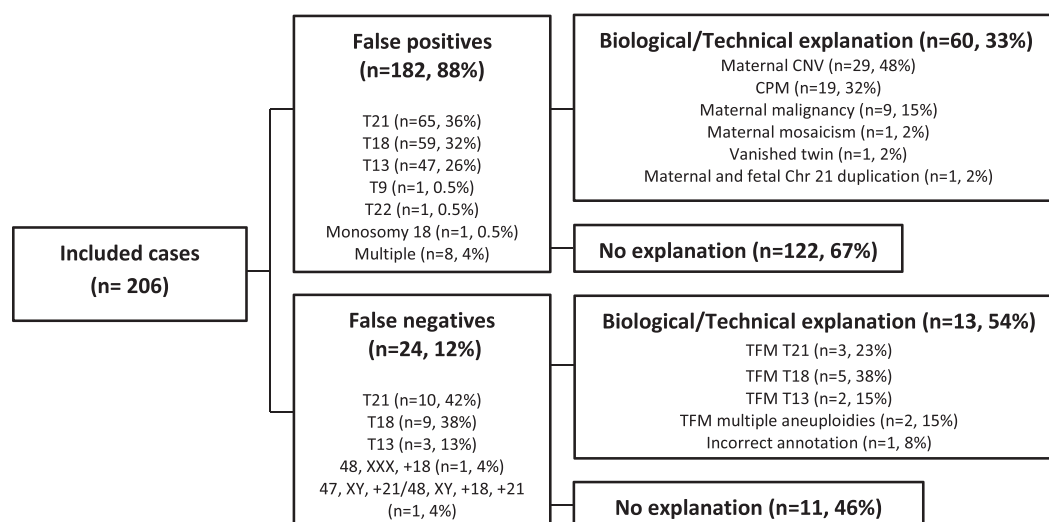


Figure 1 Flow chart to summarize included cases and the types and frequencies of each category. FP, false positive; FN, false negative; CPM, confined placental mosaicism; CNV, copy number variation; TFM, true fetal mosaicism

discordance between NIPT result and fetal sex or RhD status caused by a vanished twin have also been published.^{37,38} Although, we still need large systematic studies to fully understand the persistence of cfDNA from a vanished twin in maternal circulation.

MATERNAL COPY NUMBER VARIATIONS

In two of the included studies maternal CNV was found to be the explanation for false positive results.^{12,14} The mechanism behind maternal CNV causing false positive NIPT results was described by Snyder *et al.* in 2015.¹⁴ They evaluated the frequency and size of CNVs in a big European cohort of 19 584 persons and found that the burden of non-pathogenic CNV increases on chromosome 21, 18 and 13. In continuation hereof, they estimated that maternal CNV may contribute substantially to an elevated risk of false positive results. They evaluated four false positive cases of NIPT and identified large maternal CNV's on chromosome 18 as plausible causes of the discordant results. Although, as responded by various research groups, the influence of maternal CNV on false positive rate (FPR) is controversial and probably it depends on the utilized technology. Methods for reducing the impact of CNV on FPR have been described, as for example digital analysis of selected regions (DANSR)³⁹ and compensation algorithms,^{40,41} resulting in elimination of false positive results caused by maternal CNV.

MATERNAL MALIGNANCY

Potential biological findings for discordance between the NIPT result and the fetal karyotype include maternal malignancy, although cancer during pregnancy is a relatively rare condition with an incidence of about 1 per 1000 pregnancies.⁴² In this review, 15% of the explained false positive cases are caused by maternal malignancy, but this result does not obviously reflect the true clinical frequency.

In 2015, Bianchi *et al.* published their findings showing a relation between discordant NIPT results and maternal cancer.¹¹ Seven of the ten reported cases of maternal cancer

had an NIPT result showing multiple aneuploidies. Among the study population of 125 426 women, 39 cases had an NIPT result showing multiple aneuploidy of which seven cases (18%, 95% CI: 7.5–33.5%) were women with an occult cancer. Although an NIPT result with multiple aneuploidies caused by maternal cancer is a rare event seen in approximately one per 10 000 NIPT analyses,¹¹ Snyder *et al.* recently did follow up of multiple aneuploidies and single monosomies detected by NIPT. Among 37 women with an NIPT result of multiple aneuploidies, 19% turned out to have maternal malignancy.⁴³ Discordant NIPT results in cases with maternal malignancy can probably be explained by cell-free DNA from apoptotic malignant cells being released into maternal circulation. Bettegowda *et al.* have shown that abnormal cell-free DNA patterns are found in 80% of patients with a known metastatic solid tumour and in 50% of patients with a known localized cancer.⁴⁴

FETAL FRACTION

In 28 (14%) of the reviewed cases, the FF was not reported, and perhaps the lab did not calculate this important quality measure. In three of the unexplained false negative cases, the FF was not reported, then why an unidentified low FF might be the reason for the false negative result, especially in the two cases without a reported gestational age.

The definition of the FF of circulating cell-free DNA in maternal blood is the amount of cfDNA divided by the amount of total cell-free DNA. As FF is a ratio, it is a dynamic value that can be influenced by several factors.⁴⁵ Still some laboratories do not measure the FF, but instead they are increasing the sequencing depth to approximate claimed test performance. However, the increase in sequencing depth implies an increase of utilization of next generation sequencing reagents and consumables, resulting in an increased cost per test sample.

The minimum threshold for FF has been discussed previously. A cut-off for FF has been set to 4% by using a

statistical model based on the requirement for sufficient sequencing read depth as a function of FF.⁴⁶ Others have shown that reducing the cut-off for FF from 4% to 2%, using their specific NIPT approach rather than a theoretically determined fixed FF cut-off, a sixfold increased incidence of aneuploidy was observed in samples with $2\% < \text{FF} < 4\%$ versus samples with $\text{FF} > 4\%$.⁴⁷ However, as replied by Lüthgens *et al.*,⁴⁸ the included cohort described previously consists of only 25 aneuploidy cases with an FF below 4%, which is not enough to generate a robust validation study for the limit of detection.

Methods for calculating FF differ between laboratories, and the FF is still not always being calculated. Several methods have been described for determining FF. We are aware of six different methods based on (1) methylation,⁴⁹ (2) single nucleotide polymorphisms,³⁰ (3) quantitative polymerase chain reaction on Y-chromosome or Y-fraction calculated by DEFRAG (only for male fetuses),⁵⁰ (4) autosomal regional read counts calculated by SeqFF,⁵¹ (5) differences in length between cfDNA and maternal cfDNA⁵¹ and (6) single reads nucleosome-based FF (SANEFALCON).⁵² It is questionable whether the methods, and hence the FFs reported in this review, are comparable. We have not been able to identify studies concerning the reproducibility of FF between laboratories. It would be interesting to investigate the variation in results of FF analysed in different laboratories and with different methods. Furthermore, it seems important to generate an international standardization of FF, ensuring comparability and consistency between laboratories.

The FF is related to maternal weight and body mass index.^{22,45} In obese women, there is an increased turnover of adipocytes, thereby increasing the amount of maternal cell-free DNA and decreasing the cfDNA fraction.⁵³ According to Canick *et al.*, the reduction in cfDNA between a 110 kg and 45 kg woman is 49% while the reduction in maternal cell-free DNA between a 110 kg and a 45 kg woman is only 26%.²² The total blood volume increases with maternal weight causing a decreasing FF because of the diluting effect.⁵⁴ In laboratories not calculating the FF, maternal obesity might be a reason for false negative NIPT results.

Furthermore, gestational age influences on FF. In 2013, Wang *et al.* investigated the effects of gestational age and maternal weight among 22 384 singleton pregnancies.⁵⁴ They found that levels of cfDNA increase with a 0.1% per week between 10 and 21 weeks of gestation. Beyond 21 weeks of gestation, cfDNA increased 1% per week. CfDNA fraction was proportional to gestational age and inversely proportional to maternal weight.

OTHER FACTORS AFFECTING PERFORMANCE OF NIPT

It is well known that first and second trimester maternal serum marker levels are influenced by assisted reproductive technologies (ART) compared with naturally conceived pregnancies.^{55–57} Both maternal serum markers and cfDNA originate from placenta, why it has been hypothesized that cfDNA levels may also be affected in ART pregnancies. A few studies have tested this hypothesis by evaluating the level of cfDNA in ART and naturally conceived pregnancies.

Lambert-Messerlian *et al.*⁵⁸ found no difference in the level of cfDNA in pregnancies achieved using ART compared with naturally conceived controls. Although they identified a small but statistically significant reduction of Z-scores among euploid ART pregnancies. Recently, Sarno *et al.* demonstrated that in a mixed population of singleton and twin pregnancies the FF was significantly lower in IVF pregnancies than in naturally conceived pregnancies.⁵⁹ Also, they found a higher test failure rate in IVF pregnancies, both singleton and twin, compared with those spontaneously conceived. This implies that those methods that do not measure the FF have a higher false negative rate in IVF pregnancies. Additionally, the higher prevalence of vanishing twins in ART pregnancies should be noticed when evaluating an NIPT result of an ART pregnancy.

Only a few papers describe the technical details of their false positive and false negative cases. The NIPT methodology is complicated with numerous steps, and partial results before the final result can be reported. Consequently, working close together and sharing experiences of NIPT between laboratories is important to improve and optimize the method. Methodology and technological platforms must be described when reporting NIPT results, likewise technical details as for example Z-scores, reference intervals, FF, numbers of reads and no-call rate are essential information when comparing and exchanging NIPT results. None of the included papers reported the sequencing depth, which is an important detail, especially from labs not calculating the FF. Grati *et al.* recently demonstrated that the type of failure in no calls also influences the overall test performance, why no-call rates for technical and other reasons should also be fundamental information from NIPT providers.⁶⁰

The NIPT methodology is to a great extent based on bioinformatics, which also represents a future perspective of the test. Continuous research in bioinformatics will expand the opportunities of NIPT, and probably, we have only seen a minor part of the applicability of the technique. Applying a refined bioinformatic pipeline to identify also sub-chromosomal abnormalities has been demonstrated and is already offered from a list of commercial providers,⁶¹ although with a lower sensitivity and higher FPR than for aneuploidies. Different bioinformatics approaches also represent a source of possible discordance in results between labs, as the software platform varies and the transparency is low. The list of available software solutions for NIPT analysis and determination of FF is long and the external quality assurance non-existent. Laboratories performing NIPT should preferably be assigned to an international quality assurance organization to ensure consistent quality control. Also, establishment of an international registry for systematic recording of discordant NIPT results and their details would be advisable.

CONCLUSION

In this review of false positive and false negative NIPT cases CPM, TFM, maternal CNV or mosaicism, maternal malignancy, vanishing twin and technical or human errors was found to be possible reasons for discordance between NIPT results and

fetal karyotype. Also, low FF might be a reason for false negative cases if the lab does not measure this important parameter. Cases included in this review only represents a small part of the true number of false positive and false negative NIPT cases identified in fetal medicine clinics around the world. Furthermore, the case reporting is influenced by publication bias.

To ensure knowledge and transparency of NIPT across borders and laboratories, we suggest an establishment of an international quality assurance organization. This is important also because NIPT will develop and within a short time be expanded beyond chromosome 21, 18, 13, X and Y. A review of false positive and false negative cases, as we hereby present, illustrates the complexity of non-invasive prenatal screening. Cell-free DNA can derive from multiple sources why NIPT reflects not only the fetal karyotype but potentially also conditions as CPM, maternal cancer or a previously unrecognized maternal genetic condition. The methodology is advanced, and potential pitfalls are numerous.

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WHAT'S ALREADY KNOWN ABOUT THIS TOPIC?

- With a high sensitivity and specificity, NIPT is an incomparable prenatal screening test for trisomies 21, 18 and 13. However, the test is not diagnostic, and discordant results occur. Reasons for false positive and false negative cases are known, but not fully clarified.

WHAT DOES THIS STUDY ADD?

- Mosaicism, maternal copy number variations (CNV), maternal malignancy, vanishing twin and technical, bioinformatics or human errors were found to be reasons for discordance between NIPT result and fetal karyotype. The main part of reported cases gives no obvious explanation for the discordant results. Cases included in this paper are only a small part of the true number of discordant NIPT cases identified in fetal medicine clinics around the world, as only a small part has been published. This review gives an overview of reasons for false positive and false negative NIPT results and emphasizes the need for transparency and knowledge exchange between clinics and laboratories.

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